Inductively-Defined Sets

An ***inductive definition of a set*** has three components (clauses):

1. The ***basis clause*** establishes that certain objects are in the set.

   These elements constitute the "building blocks" for constructing further elements in the set.

   For example: 0 *is a basic natural number.*

2. The ***inductive clause*** defines the ways in which elements of a set can be used to produce further elements which are also in the set.

   For example: if n *is a natural number,*
   then so is $n+1$*.*

3. The ***extremal clause*** asserts that no object is an element of the set being defined unless it can be established from the first two clauses.

$\mathbb{N}$: An Example Inductively-Defined Set

The set of natural numbers $\mathbb{N}$ can be defined as follows:

1. $0 \in \mathbb{N}$.
2. $(n+1) \in \mathbb{N}$ whenever $n \in \mathbb{N}$.
3. Nothing is in $\mathbb{N}$ except through clauses (1) and (2).

   That is, $\mathbb{N}$ is the smallest set that satisfies the first two clauses.

The basis clause (1) says that $0 \in \mathbb{N}$;

The inductive clause (2) says that from any natural number n we can produce another natural number $n+1$.

In this way we can conclude that all values $0, 1, 2, 3, \ldots$ are in $\mathbb{N}$:

- $0 \in \mathbb{N}$ (by the basis clause);
- hence $1 = 0+1 \in \mathbb{N}$ (by the inductive clause);
- hence $2 = 1+1 \in \mathbb{N}$ (by the inductive clause);
- hence $3 = 2+1 \in \mathbb{N}$ (by the inductive clause);

  ⋮

The Extremal Clause

The extremal clause (3) above says that an element of $\mathbb{N}$ ***must*** be either 0 (from the basis clause) or the successor of (ie, one plus) another element of n.

For example, $2.5 \notin \mathbb{N}$ as we could not generate 2.5 from the first two clauses: it is not 0; and repeatedly adding 1 to 0 will not generate the value 2.5.

In other words, the extremal clause says that the set being defined is the ***smallest*** set which satisfies the first two clauses.

For example, the set of integers

$$\mathbb{Z} = \{ \ldots, -4, -3, -2, -1, 0, 1, 2, 3, 4, \ldots \}$$

satisfies clauses (1) and (2). But it is not the smallest set that does.

The set $\{0, 1, 2, 3, 4, \ldots\}$ satisfies clauses (1) and (2), and every element in this set is generated by clauses (1) and (2); hence this is the set being defined.

Inductively-Defined Subsets of $\mathbb{N}$

Example: We can inductively define the set

$$\textsc{Odd} \;=\; \{1, 3, 5, 7, \ldots\}$$

of odd natural numbers as the smallest set satisfying the following:

1. $1 \in \textsc{Odd}$; and
2. if $n \in \textsc{Odd}$ then $(n+2) \in \textsc{Odd}$.

Note how the extremal clause is incorporated into the preamble of the definition.

In particular, the set $\mathbb{N}$ of natural numbers satisfies the two clauses in this definition, but is not the smallest set satisfying these clauses, so it is not the set being defined.

Inductively-Defined Subset of $\mathbb{N}$

Exercise: Give an inductive definition for the set:

$$\textsc{PowersOf2} = \{1, 2, 4, 8, 16, 32, 64, 128, \ldots\}.$$

Exercise: Give an inductive definition for the set:

$$X = \{1, 2, 5, 10, 21, 42, 85, 170, 341, 682, 1\,365, 2\,730, 5\,461, 10\,922, \ldots\}.$$

Inductively-Defined Subset of $\mathbb{N}$

Exercise What is the smallest set $X \subseteq \mathbb{N}$ such that:

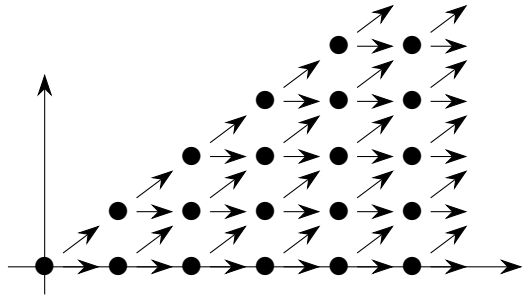
1. $2 \in X$; and
2. if $n \in X$ then $(n+3) \in X$ and $(n+5) \in X$

Exercise What is the smallest set $X \subseteq \mathbb{N}$ such that:

1. $1 \in X$ and $3 \in X$; and
2. if $n \in X$ then $(n \times 2) \in X$ and $(n+5) \in X$.

Inductively-Defined Subset of $\mathbb{N}\times\mathbb{N}$

Example: $W = \{(x,y) \in \mathbb{N} : x \geq y\}$



There are different ways to define W inductively.

For example, it could be defined to be the smallest set satisfying any one of the three pairs of clauses.

A. W is the smallest set satisfying the following:

1. $(0,0) \in W$; and
2. If $(i,j) \in W$ then $(i+1,j) \in W$ and $(i+1,j+1) \in W$.

B. W is the smallest set satisfying the following:

1. $(i,0) \in W$ for each $i \in \mathbb{N}$; and
2. If $(i,j) \in W$ then $(i+1,j+1) \in W$.

C. W is the smallest set satisfying the following:

1. $(i,i) \in W$ for each $i \in \mathbb{N}$; and
2. If $(i,j) \in W$ then $(i+1,j) \in W$.

Example: Finite Powersets

Given a finite set S, the powerset $\mathcal{P}(S)$ of S is the smallest set satisfying:

1. $\emptyset \in \mathcal{P}(S)$; and
2. If $X \in \mathcal{P}(S)$ and $a \in S$ then $X \cup \{a\} \in \mathcal{P}(S)$.

For example, if $S = \{1, 2\}$, then by the basis clause, $\emptyset \in \mathcal{P}(S)$; and thus by one application of the inductive clause we get that the following sets are in $\mathcal{P}(S)$:

$$\emptyset \cup \{1\} = \{1\} \quad \text{and} \quad \emptyset \cup \{2\} = \{2\}.$$

A second application of the induction clause puts the following sets in $\mathcal{P}(S)$:

$$\begin{array}{lll} \emptyset \cup \{1\} = \{1\} & \{1\} \cup \{1\} = \{1\} & \{2\} \cup \{1\} = \{1, 2\} \\ \emptyset \cup \{2\} = \{2\} & \{1\} \cup \{2\} = \{1, 2\} & \{2\} \cup \{2\} = \{2\} \end{array}$$

A further application of the inductive clause would not add any new elements. Thus $\mathcal{P}(\{1, 2\}) = \{\emptyset, \{1\}, \{2\}, \{1, 2\}\}$.

Alphabets and Words

Given a set A of letters (ie, an ***alphabet***), A^* represents the set of ***words*** (ie, arbitrary sequences of letters) that can be formed from letters taken from A, including the ***empty word*** ε consisting of no letters.

For example, if $A = \{a, b\}$ then the following are all words in A^*:

$$\varepsilon,\ a,\ b,\ aa,\ ab,\ ba,\ bb,$$
$$aaa,\ aab,\ aba, abb, baa, bab, bba, bbb,$$
$$\vdots$$
$$\ldots, bbabbaba,\ \ldots$$
$$\vdots$$

The set A^* of words over an alphabet A is defined inductively as the smallest set satisfying the following:

1. $\varepsilon \in A^*$; and
2. if $w \in A^*$ and $a \in A$ then $aw \in A^*$.

Inductively-Defined Sets of Words

Exercise: What is the smallest set $X \subseteq \{ a, b \}^*$ such that:

1. $\varepsilon \in X$; and
2. if $w \in X$ then $awb \in X$.

Exercise: What is the smallest set $Y \subseteq \{ a, b \}^*$ such that:

1. $\{\varepsilon, a, b\} \subseteq Y$; and
2. if $s \in Y$ then $\{asa, bsb\} \subseteq Y$.

BNF – Backus-Naur Form

A succinct way to present the inductive definition of the set A^* is as follows (so-called ***BNF notation***):

$$w \quad ::= \quad \varepsilon \mid aw \qquad \text{(where } a \in A\text{)}$$

BNF notation is very useful for presenting the syntax of logics and languages.

The set of ***nonempty words*** $A^+ = A^* \setminus \{\varepsilon\}$ can be defined by:

1. $a \in A^+$ for every $a \in A$; and
2. if $w \in A^+$ and $a \in A$ then $aw \in A^+$.

or in BNF by:

$$w \quad ::= \quad a \mid aw \qquad \text{(where } a \in A\text{)}$$

For example, taking the alphabet D of decimal digits:

$$D = \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$$

D^+ is the set of all decimal numbers (including those with leading zeros).

BNF for Logic Formulas

The set of propositional formulas is defined inductively as the smallest set satisfying the following:

1. *true* and *false*, and propositional variables are propositional formulas.
2. If p and q are propositional formulas then so are $\neg p$, $p \vee q$, $p \wedge q$, $p \Rightarrow q$ and $p \Leftrightarrow q$.

This is represented by the following BNF equation:

$$p, q \quad ::= \quad \textit{true} \mid \textit{false} \mid P \mid \neg p \mid p \vee q \mid p \wedge q \mid p \Rightarrow q \mid p \Leftrightarrow q$$

Here, P ranges over propositional variables.

The set of predicate logic formulas is given by the following BNF equation:

$$\begin{aligned} p, q \quad ::= \quad & \textit{true} \mid \textit{false} \mid P(x_1, \ldots, x_n) \mid \forall x\, p \mid \exists x\, p \\ & \mid \neg p \mid p \vee q \mid p \wedge q \mid p \Rightarrow q \mid p \Leftrightarrow q \end{aligned}$$

Here, $P(x_1, \ldots, x_n)$ ranges over atomic predicates, and x ranges over variables.